

ISBA 2411 | FINAL TEAM PROJECT

MSBA Job Matching

Grounded NLP decision support
for career advisors

Search | Fit | Evidence | Human review

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The advisor must separate fit, uncertainty, and risk

A useful system improves a specific decision: forward, hold, or investigate.

FORWARD

Strong entry-level analytics fit supported by title, skills, and location evidence.

HOLD

Senior, engineering-heavy, or clearly misaligned roles require a conservative screen.

INVESTIGATE

Incomplete fields or missing authorization language require human follow-up.

Role fit and work authorization are related in workflow, but they are not the same prediction task.

Label quality, not data volume, is the bottleneck

The final model uses the full training split and audits all 785,741 source rows.

785,741

original public job
records

100,000

stratified project
sample

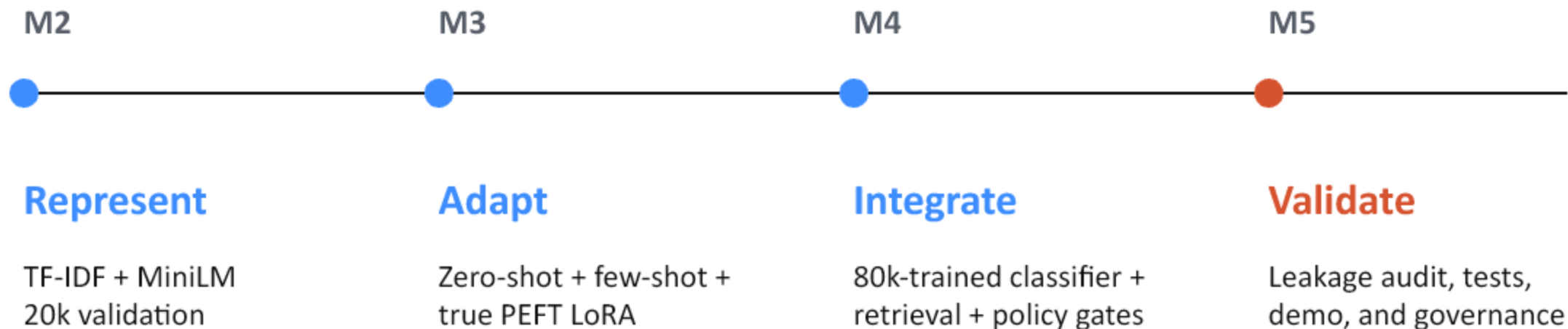
77,135

deduplicated training
rows

230

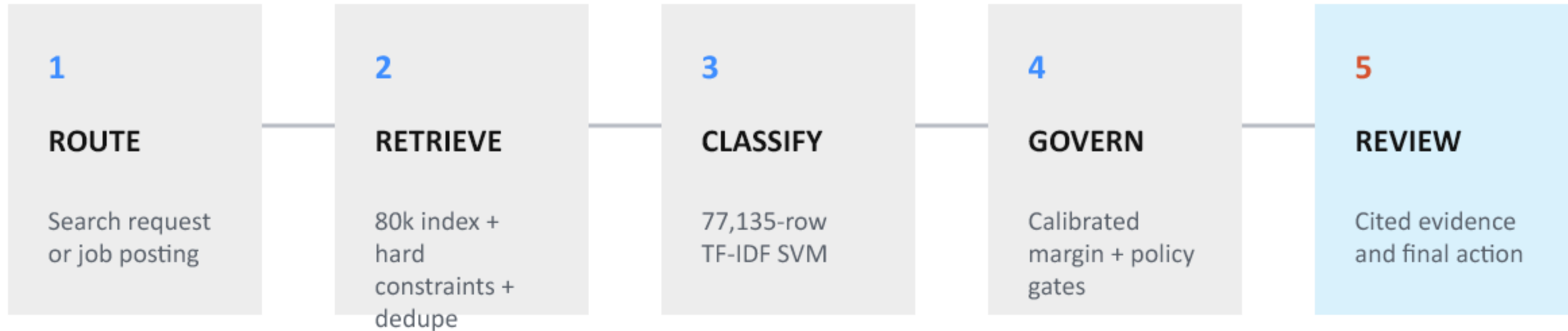
unique authorization
candidates

Each milestone changed the final technical decision



Two input modes prevent a subtle but costly task error

Search requests retrieve candidates; only job postings receive a role-fit label.



A high role-fit score can never override an explicit authorization restriction.

A real search returns constrained, inspectable evidence

```
PS> python -m jobmatch.demo --recording
```

```
=====
MSBA JOB MATCHING | FINAL CAPSTONE | REAL END-TO-END RUN
=====
SCENE 4 OF 9 | REAL OUTPUT 1: GROUNDED SEARCH RESULTS
Search text is not misclassified as if it were a job posting.
-----
Input mode           : search
Posting label        : not_applicable_search_query
Country constraint   : United States
Entry-level constraint: True
Requested skills     : sql, python, excel, tableau, power bi
Advisor review       : True

TOP RETRIEVED POSTINGS FROM THE REAL 80,000-ROW INDEX
#  Role title           Company           Location
1  Data Analyst         HireMatch        Anywhere
2  Data Analyst         HMP Global       Anywhere
3  Data Analyst         AMIRI            Los Angeles, CA

Grounding caveat: authorization evidence is not available in the public source.

This screen remains visible for review.
```

6 audit queries -> 36 results -> 0 hard-constraint violations -> 0 duplicate company-title pairs

High role fit never overrides an explicit restriction

```
PS> python -m jobmatch.demo --recording
```

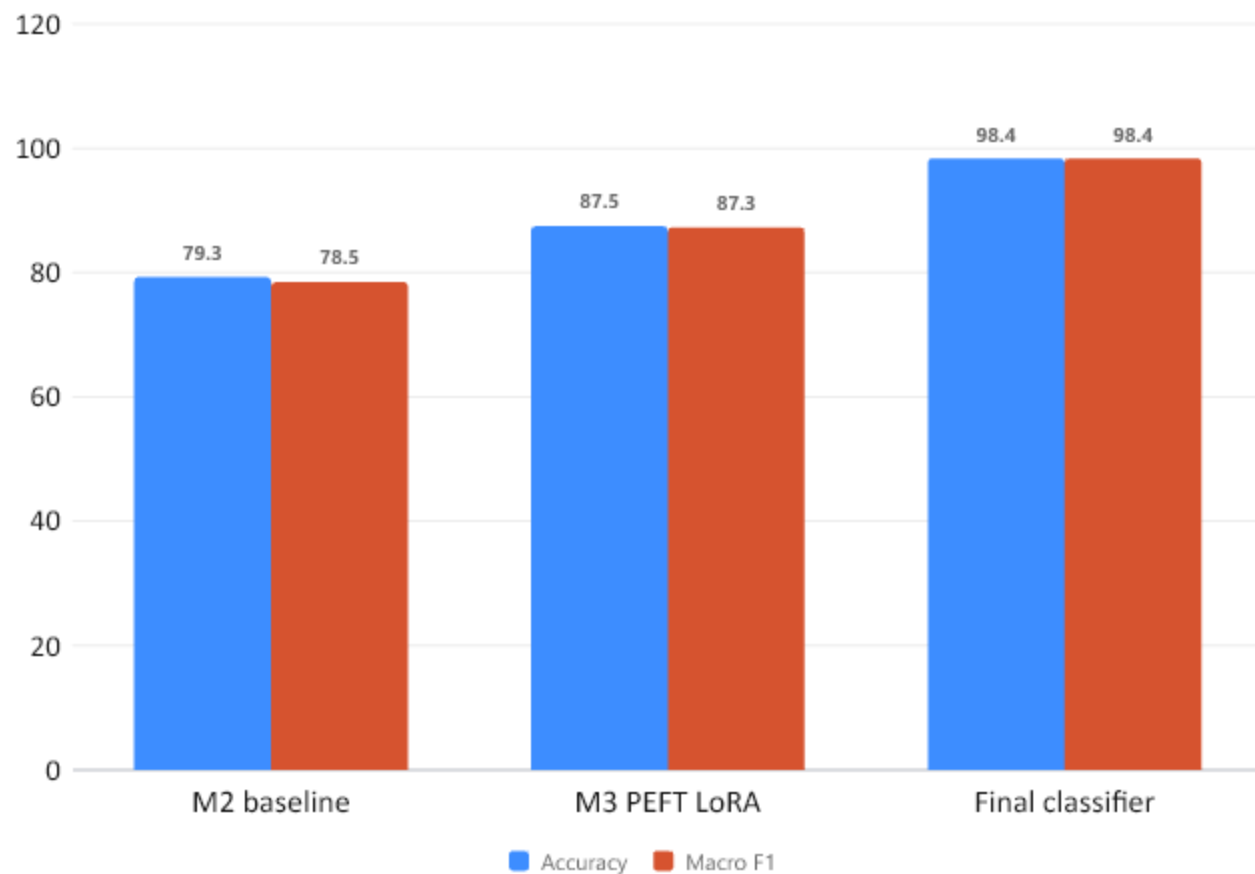
```
=====
MSBA JOB MATCHING | FINAL CAPSTONE | REAL END-TO-END RUN
=====
SCENE 6 OF 9 | REAL OUTPUT 2: ROLE FIT CANNOT OVERRIDE POLICY
The model prediction remains visible, while the governance gate controls the action.
-----
Model role-fit label   : high_fit
Prediction margin      : 4.422
Authorization evidence: explicit source text: no OPT/CPT
Review required        : True
Recommended action     : Hold - explicit authorization restriction requires advisor verification
Policy reason          : explicit authorization restriction requires review

EDGE-CASE RESULT
High role fit is preserved, but the workflow action is HOLD for advisor verification.

This screen remains visible for review.
```

No OPT/CPT -> HOLD. Missing core fields -> UNCLEAR + REVIEW.

More data worked; extra complexity did not



98.4% fixed-set macro F1

97.9% novel-text macro F1

0 unsupported
authorization claims

The final 3.8 MB model trains in under one minute.

The remaining risk is label validity, not model capacity

FIXED 4K ERRORS BY TRUE LABEL



DEPLOYMENT RULE

Treat 0.984 as weak-label agreement, not advisor accuracy.

Operational validation:
91.6% selective coverage
99.47% retained accuracy
36 search results, 0 violations

Ready for a supervised Career Center pilot

Deploy with guardrails

Prioritize, do not auto-reject
Keep advisor confirmation
Measure time saved and overrides

```
MSBA Job Matching LIVE PYTHON RUN

PS> python -m jobmatch.demo --recording

-----
MSBA JOB MATCHING | FINAL CAPSTONE | REAL END-TO-END RUN
-----
SCENE 7 OF 9 | REAL BATCH REVIEW WORKFLOW
The same production system now processes a three-row CSV queue.
-----
INPUT : demo/batch.csv
OUTPUT: outputs/batch.csv

Executing process_batch_frame(...) now ...
Completed 3 rows in 0.06 seconds.

Row  Mode   Label   Review Recommended action
1  search  not_appl... true  Advisor review before forwarding
2  posting high fil  True   Hold - explicit authorization restricti...
3  posting unclear  True   Investigate - advisor review required

Each output row preserves policy reasons, evidence, constraints, and the full JSON result.
This screen remains visible for review.

Actual stdout | 100,000-row project data | trained model 02:20
```

98.4% = weak-label agreement

Next: label the 1,000-row advisor queue and measure real workflow value.